BEAST 3: What's New for Package Developers

BEAST Core Developer Team

2026

Outline

1. Why Maven + JPMS?
2. Build & Tooling
3. Strongly Typed Spec System
4. New Distributions & Operators
5. API Removals & Breaking Changes
6. Package System
7. Testing & Dependencies
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The Big Picture

Three pillars of change in BEAST 3:

1. **Maven + JPMS**: modern build system and module isolation
2. **Strongly-typed spec system**: compile-time checked inputs and domain types
3. **New package distribution**: Maven Central JARs, module-based service discovery

Goal

Catch more errors at compile time, simplify dependency management, and make packages first-class citizens in the Java module system.

Problems We Wanted to Solve

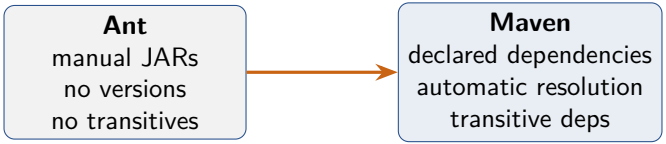
Build & dependencies

- Ant `build.xml`: manual JAR management, no transitive dependency resolution
- Adding a library means hand-copying JARs and editing classpaths
- No standard way to publish or consume packages

Classloading & isolation

- Custom classloader predates Java modules by 15+ years
- Source of "magic" that confuses new developers and breaks standard tooling
- Two packages bundling different versions of the same library can conflict

Why Maven? Dependency Management



- Declare what you need in `pom.xml`; Maven fetches, caches, and versions it
- Transitive dependencies resolved automatically: no more "which JARs do I need?"
- Reproducible builds: everyone gets the same dependency tree

Why Maven? Ecosystem & Tooling

- **IDE integration:** IntelliJ and Eclipse understand Maven projects natively: import, build, run, debug with zero configuration
- **CI/CD:** GitHub Actions, Jenkins, etc. all have first-class Maven support
- **Publishing:** packages can be published to Maven Central: the standard Java package registry, used by millions of projects
- **AI-assisted development:** Copilot, agentic coding tools understand Maven projects and JPMS modules; they don't understand bespoke build systems
- **Documentation & community:** vast ecosystem of tutorials, StackOverflow answers, and plugins

Moving to Maven means BEAST developers benefit from the entire Java ecosystem's tooling instead of maintaining our own.

Why JPMS Modules?

The **Java Platform Module System** (introduced in Java 9) gives us:

1. **Strong encapsulation**: each module declares what it exports; internal classes are truly hidden
2. **Reliable configuration**: missing dependencies are caught at startup, not at runtime via `ClassNotFoundException`
3. **Per-package isolation**: each BEAST package gets its own `ModuleLayer`; two packages can bundle different versions of the same library without conflict
4. **No split packages**: JPMS enforces that each Java package belongs to exactly one module, so good development practices are guaranteed rather than hoped for
5. **Service discovery**: `module-info.java` replaces classpath scanning and `version.xml` as the primary way to register BEASTObjects

JPMS replaces our custom classloader with a standard Java mechanism that works with, not against, the platform.

What Was Wrong with the Custom Classloader?

- Using BEAST as a **library** in other projects (e.g. LPhyBEAST) was nearly impossible: the custom classloader conflicts with the host application's class loading
- Debugging class-not-found errors required **BEAST-specific knowledge** that isn't transferable to other Java projects
- New developers and contributors faced a steep **onboarding barrier**

BEAST 3 fix: JPMS `ModuleLayer` provides per-package isolation out of the box. Standard Java, standard tooling, no magic.

Ant → Maven

BEAST 2

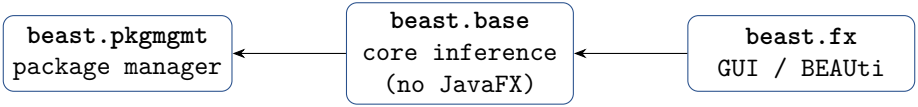
- Ant `build.xml`
- Manual JAR management
- One monolithic project
- Classpath-based

BEAST 3

- Maven `pom.xml`
- Automatic dependency resolution
- Multi-module project
- Module-path-based (JPMS)

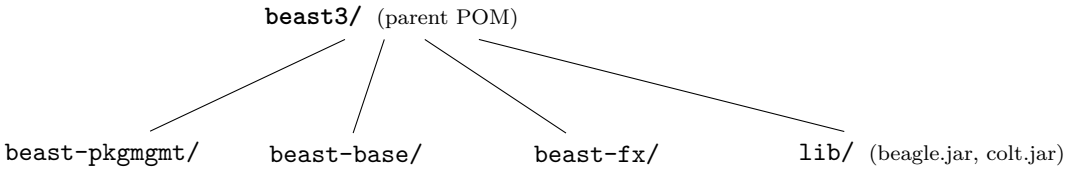
Java 25 + JPMS Modules

- Requires **JDK 25** (was Java 8/11)
- JavaFX is a **Maven dependency**: no bundled JDK needed
- Three JPMS modules, each with `module-info.java`:



All declared as `open module` (reflection OK for XML unmarshalling).

Project Structure



- `beast-base` has **no JavaFX dependency**: can be used headlessly
- `beagle` and `colt` ship as modular JARs in `lib/`

ModuleLayers replace custom classloader

The custom classloader was an idiosyncratic barrier

BEAST 2's plugin classloader predates Java's module system by 15+ years. It was the primary source of "magic" that confused new developers and defeated standard tooling.

- Using BEAST as a **library** in other projects (e.g. LPhyBEAST) was nearly impossible because the custom classloader conflicts with the host application's class loading
- AI-assisted tools (Copilot, agentic coding agents) don't understand bespoke classloaders, but they do understand Maven projects and JPMS modules
- Debugging class-not-found errors requires BEAST-specific knowledge that isn't transferable

BEAST 3 fix: JPMS `ModuleLayer` does a better job out of the box, with per-package isolation that eliminates class conflicts the custom classloader can't deal with. Standard Java, standard tooling, no magic.

What This Means for Your Package

1. **Recommended:** create a Maven `pom.xml` for your package
   - See `beast-package-skeleton` on GitHub
2. **Alternative:** keep Ant, but update classpath JARs
3. Add a `module-info.java` with `provides` declarations
4. Target **Java 25** (use `-release 25`)

Tip

The `beast-package-skeleton` repo is a ready-to-copy Maven project with all of this pre-configured.

Why Types?

Problem: In BEAST 2, almost everything is `Input<RealParameter>` or `Input<Function>`

- Type errors discovered at **runtime** (wrong parameter passed to wrong input)
- Bounds (`lower="0"`) are stringly-typed XML attributes
- No way to express "this must be a positive real" in the Java API

Solution: New `beast.base.spec` hierarchy

- Compile-time checked typed inputs
- Domain types encode valid ranges
- Self-documenting model structure

Parameter Types

`RealParameter` is replaced by specific types:

BEAST 2	**BEAST 3**	**Use for**
`RealParameter` (dim=1)	`RealScalarParam`	single real value
`RealParameter` (dim>1)	`RealVectorParam`	vector of reals
`IntegerParameter`	`IntScalarParam` / `IntVectorParam`	integer values
`BooleanParameter`	`BoolScalarParam` / `BoolVectorParam`	boolean values
`RealParameter` (simplex)	`SimplexParam`	probability simplex

Domain Types

Domains replace explicit `lower`/`upper` bounds:

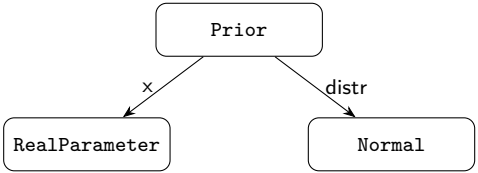
BEAST 2 XML pattern	**BEAST 3 domain**
`RealParameter lower="0"`	`PositiveReal.INSTANCE`
`RealParameter lower="0" upper="1"`	`UnitInterval.INSTANCE`
`RealParameter` (unbounded)	`Real.INSTANCE`
`IntegerParameter lower="0"`	`NonNegativeInt.INSTANCE`
`IntegerParameter lower="1"`	`PositiveInt.INSTANCE`

Domains propagate through the model graph at initialization time.

"Distribution IS the Prior"

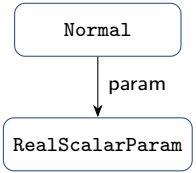
BEAST 2

`Prior` wraps distribution + parameter:



BEAST 3

Distribution **is** the prior:



No separate `Prior` wrapper needed: each distribution has a `param` input directly (from `TensorDistribution`).

Spec Classes Coexist with Legacy

- Legacy classes are **deprecated**, not removed
- Each deprecated class links to its spec replacement:

```
/**
 * @deprecated use {@link RealScalarParam} or {@link RealVectorParam}
 */
public class RealParameter extends ...
```

- Your IDE shows deprecated usage (strikethrough)
- **Migrate incrementally**: old and new classes can coexist
- Consider a `yourpackage.spec` sub-package for backward compatibility

Before / After: Input Declaration

BEAST 2

```
public Input<RealParameter> kappaInput =
    new Input<>("kappa",
    "ts/tv ratio",
    Validate.REQUIRED);
```

BEAST 3

```
public Input<RealScalar<PositiveReal>>
    kappaInput = new Input<>("kappa",
    "ts/tv ratio",
    Validate.REQUIRED);
```

- Domain (`PositiveReal`) is part of the type: no runtime `lower="0"` needed
- Compiler catches if someone passes a `UnitInterval` parameter where `PositiveReal` is expected

When NOT to Use Spec Types

Spec parameters are for MCMC-estimated quantities only.

Keep plain `Input<Integer>`, `Input<Double>`, etc. for:

- Dimension counts
- Category numbers
- Boolean flags
- Configuration / initializer values

Rule of thumb

If it appears in the MCMC state, use a spec parameter type.
If it's fixed configuration, use a plain `Input<>`.

New Distributions

Genuinely new: not present in BEAST 2:

Continuous

- `Cauchy`: location / scale
- `GammaMean`: mean parameterization

Discrete & Wrappers

- `Bernoulli`: boolean with prob p
- `IntUniform`: discrete uniform
- `IID`: scalar dist $\rightarrow$ vector
- `OffsetReal`: additive offset
- `TruncatedReal`: truncation bounds

New / Refactored Operators

- **`IntervalScaleOperator`** (new): scales intervals between consecutive node heights; preserves topology, changes relative branch lengths
- **`ScaleTreeOperator`** (split): dedicated tree-scaling operator, separated from the parameter `ScaleOperator`
- **Spec `UpDownOperator`**: `up`/`down` take `List<Scalable>`; uses Bactrian kernel by default

Simplified Default Tree Operator Set

BEAST 2 (BICEPS)

- `EpochFlexOperator` (top)
- `EpochFlexOperator` (all)
- `TreeStretchOperator`
- `BactrianScaleOperator`
- `BactrianNodeOperator`
- `BactrianSubtreeSlide`
- `Exchange` (narrow/wide)
- `WilsonBalding`

BEAST 3

- **`UpDownOperator`** (replaces 3 BICEPS ops)
- `BactrianScaleOperator`
- `BactrianNodeOperator`
- `BactrianSubtreeSlide`
- `Exchange` (narrow/wide)
- `WilsonBalding`

StateNode Changes

Breaking

`StateNode` is **no longer a `Function`**.

- If your code relies on `StateNode` implementing `Function`, explicitly implement `Function` yourself (intermediate step) or migrate to typed spec parameters

Breaking

`StateNode.scale()` has been **removed**.

- Implement the `Scalable` interface if your `StateNode` needs scaling support

Removed Classes

Removed	Replacement
`beast.base.inference.Evaluator`	(removed; impacted MCMC subclasses)
`AscertainedAlignment`	Use standard `Alignment`

- Check your code for imports of these classes
- `Evaluator` mainly affected custom MCMC acceptance logic

Narrowed Operator Input Types

Spec operators accept **typed** parameters:

- `ScaleOperator.parameter` → `Scalable` interface
- `RealRandomWalkOperator` → `RealVectorParam` xor `RealScalarParam`
- `BitFlipOperator` → `BoolVectorParam`
- `DeltaExchangeOperator`: adds typed inputs (`rvparameter`, `ivparameter`, etc.)

See the migration guide for the complete operator mapping table.

Package System Design Principles

1. **One ModuleLayer per package:** namespace isolation without losing visibility to the core API. Two packages can bundle different versions of the same library without conflict.
2. **Same code path for dev and deployment:** `module-info.java` `provides` declarations are the single source of truth. A developer in IntelliJ and an end user with an installed ZIP see the same service discovery.
3. **Graceful degradation:** modules with unsatisfied dependencies (e.g. FX modules on a headless cluster) are silently skipped. Core modules still load.
4. **First-found wins:** when the same JPMS module appears in multiple locations (boot layer, installed package, archive), the first one loaded wins. No silent duplicates, no "reads more than one module" errors.
5. **Dual distribution:** ZIP archives (CBAN) and Maven Central JARs converge on the same ModuleLayer creation path.

Development vs Deployment

Development (IDE)

- All modules in the **boot layer**
- IDE resolves module path from Maven
- Package manager has nothing to do
- Services discovered via `module-info.java`
- Debug, hot-reload as normal

Deployment (user install)

- Only core modules in boot layer
- External packages loaded into child **ModuleLayers** at startup
- Two storage formats coexist
- Same service discovery mechanism

Service Discovery: `module-info.java`

Primary mechanism: BEAST scans module descriptors at startup.

```
open module my.beast.package {
    requires beast.pkgmgmt;
    requires beast.base;

    provides beast.base.core.BEASTInterface with
        my.beast.package.MyModel,
        my.beast.package.MyOperator;
}
```

- Works in both the boot layer (IDE development) and plugin layers (deployed packages)
- IntelliJ/Eclipse discover services automatically when both projects are open

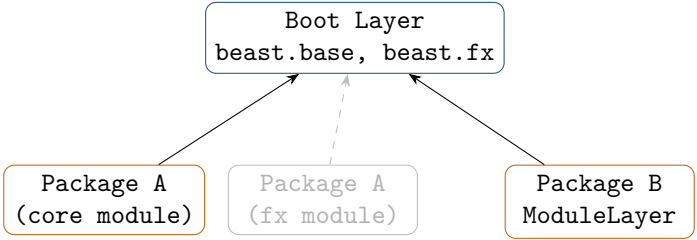
Service Discovery: `version.xml` Fallback

- **Still works** for non-modular JARs
- JARs without `module-info` are treated as automatic modules
- Services registered from `version.xml` as before
- Useful for gradual migration

Recommendation

Add `module-info.java` when you can: it is the primary mechanism and gives you compile-time module dependency checking.

Plugin ModuleLayer Isolation



- Each deployed package gets its **own ModuleLayer**
- No classpath conflicts between packages
- Modules with unsatisfied dependencies (e.g. FX on a headless cluster) are **automatically skipped**; core modules still load

Two Storage Models

Both formats live under `~/.beast/2.8/`:

Maven packages (recommended)

- `maven-packages.xml`: installed coordinates
- `maven-repo/`: local JAR cache (package + transitive deps)
- Remote repos: Maven Central + extras in `beauti.properties`
- Index needed because the cache is opaque

ZIP packages (legacy)

- One **subdirectory per package**:
  `MyPackage/lib/*.jar`
  `MyPackage/version.xml`
- Directory structure *is* the index: any subdirectory with `version.xml` is a package
- No external index needed

Why two models?

ZIP packages are self-describing (directory = package). Maven packages need an explicit index because the local cache holds transitive dependencies alongside the package itself.

Package Loading Precedence

`PackageManager.loadExternalJars()` runs when BEAST starts:

1. **Load Maven packages:** resolve coordinates from `maven-packages.xml`, create a ModuleLayer per package. Maven is the recommended format, so these take precedence.
2. **Scan package directories:** find installed ZIP packages in `~/.beast/2.8/` (or `BEAST_PACKAGE_PATH`)
3. **Create ModuleLayer per ZIP package:**
   - Discover JPMS modules from the package's JARs
   - Exclude modules already in boot layer or Maven layers (no duplicates)
   - Filter out modules whose `requires` can't be satisfied
   - `resolveAndBind()` on the remaining modules
4. **Register with `BEASTClassLoader`:** services from both `module-info.java` and `version.xml` merged into registry

Precedence: boot layer → Maven → ZIP. If the same module exists in both formats, the Maven version wins.

Maven Central Distribution

Packages can be published as plain Maven Central JARs.

- Users install via BEAUti: **Install from Maven** button
- Or CLI: `packagemanager -maven groupId:artifactId:version`
- Resolution via Apache Maven Resolver

Three components:

- `maven-packages.xml`: records installed coordinates
- `~/.beast/2.8/maven-repo/`: local JAR cache
- Remote repos: Maven Central (always) + custom repos in `beauti.properties`

- `packagemanager -addMavenRepository <url>`: add a custom repo
- ZIP packages still work alongside Maven packages

beast-package-skeleton

Ready-to-copy template project on GitHub:

- Maven `pom.xml` with BEAST 3 dependencies
- `module-info.java` with `provides` declarations
- Example `ScalarDistribution`, operator, and tests
- BEAST XML using the strongly-typed spec API
- Maven Central release profile (source, javadoc, GPG, publishing plugin)
- `version.xml` JAR embedding

Start here: `https://github.com/CompEvol/beast-package-skeleton`

Testing Upgrades

JUnit 5

- Primary framework (Jupiter 5.8.2)
- JUnit 4 still compiles via vintage engine
- `@Tag("slow")` for long MCMC tests
- `mvn test`: fast tests only
- `mvn test -Pslow-tests`: all

TestFX (GUI testing)

- BEAUti tests run **headlessly**
- Monocle Glass platform
- No display server needed
- TestFX 4.0.18 + JUnit 5 integration

Dependency Updates

Dependency	**BEAST 2**	**BEAST 3**
Java	8 / 11	25
Build	Ant	Maven 3.9+
Math	commons-math3	commons-math4-legacy 4.0 + commons-numbers / rng / statistics
ANTLR	4.x (bundled)	4.13.2 (Maven)
JavaFX	bundled with JDK	25.0.2 (Maven)
JUnit	4	5 (Jupiter 5.8.2)
GUI tests	–	TestFX 4.0.18

BEAUti & Templates

- **fxtemplates namespacing**: templates moved to module-unique paths (`beast.fx/fxtemplates/`) to avoid JPMS split-package conflicts
- **Module resource scanning**: `BeautiDoc` scans JPMS module resources, so templates are discovered from JARs on the module path
- External packages use their own namespace: e.g. `beast.morph.models.fx/fxtemplates/`
- Legacy path (`fxtemplates/*.xml`) still supported as fallback

Migration Checklist

1. **Update build**: create `pom.xml` (or update Ant classpath JARs), target Java 25
2. **Fix compile errors**: `StateNode` not `Function`, `scale()` removed, `Evaluator` gone
3. **Add `module-info.java`**: declare `provides` for your `BEASTInterface` implementations
4. **Migrate to spec types**: replace `RealParameter` inputs with `RealScalarParam` / `RealVectorParam` and domain types
5. **Update XML & templates**: use spec class names, new operator set, namespaced fxtemplates

Key Breaking Changes

Area	**BEAST 2**	**BEAST 3**
Build	Ant, Java 8/11	Maven, Java 25
Modules	Custom classloader	JPMS `ModuleLayer`
Parameters	`RealParameter`	Typed spec params + domains
Priors	`Prior` wrapper	Distribution is the prior
`StateNode`	implements `Function`	No longer a `Function`
`StateNode.scale()`	built-in	Removed; use `Scalable`
JavaFX	Bundled with JDK	Maven dependency

Legacy classes are **deprecated, not removed**: migrate incrementally.

Timeline

Now	Package migration	**Release + paper**	ISBA
Mar 2026	Apr–May	26 Jun 2026	26 Jun–5 Jul
Stakeholders meeting (19 Mar)	20+ packages across multiple groups	Before SMBE Copenhagen	

Resources

Migration guide `scripts/migration-guide.md`: class mapping tables, step-by-step instructions

CHANGELOG `CHANGELOG.md`: full list of what changed and why

Package skeleton `https://github.com/CompEvol/beast-package-skeleton`

BEAST 3 repo `https://github.com/CompEvol/beast3`